

Debiasing by obfuscating with 007-classifiers promotes fairness in multi-community settings

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Abstract

While there has been considerable amount of research on bias mitigation algorithms, two properties: multi-community perspective and fairness to *all* communities have not been given sufficient attention. Focusing on these, we propose an obfuscation based data augmentation debiasing approach. In it we add to the training data *obfuscated* versions of *all* false positive instances irrespective of source community. We test our approach by debiasing toxicity classifiers built using 5 neural models (multi layer perceptron model and masked language models) and 3 datasets in a 4 communities setting. We also explore 4 different obfuscators for debiasing. Results demonstrate the merits of our approach: bias is reduced for almost all of our runs without sacrificing false positive rates or F1 scores for minority or majority communities. In contrast, the 4 state of the art baselines typically make performance sacrifices (often large) while reducing bias. Crucially, we demonstrate that it is possible to debias while maintaining standards for both minority and majority communities.

Note: This paper contains examples of toxic texts/posts.

1 Introduction

Even though deep learning based text classifiers are popular, enthusiasm is tempered because of their biases against particular communities. There is now a definite expectation that these systems should perform well on the whole as well as for different groups; groups that may be defined using sensitive attributes such as ethnicity (our focus), gender, religion. Thus, bias assessment in text classifiers and their mitigation (debiasing) is now an active area of research. There are now four broad categories of mitigation algorithms. These are *pre-processing* - adjust the training data (Calmon et al., 2017; Ball-Burack et al., 2021; Chakraborty et al., 2021; Almuzaini et al., 2022); *in-processing* - adjust

model training (Zhang et al., 2018; Ball-Burack et al., 2021; Yazdani-Jahromi et al., 2022; Kumar et al., 2023); *intra-processing* - adjust through model fine-tuning (Savani et al., 2020) and *post-processing* - adjust predicted labels (Hardt et al., 2016; Lohia et al., 2019; Qian et al., 2021). Our algorithm is in the pre-processing category.

Limitations of prior debiasing algorithms: (1)

Prior debiasing algorithms largely consider a single community (typically African American (AE)) as minority and the rest as a single majority community. While we do not question the merits of including AE as minority we suggest that depending on model-dataset combination, communities such as Hispanic might also suffer biases. Moreover, datasets typically derive from social media platforms in which many communities engage emphasizing the need for a multi-community perspective. (2) While reducing bias, these algorithms often make sacrifices with regards to other communities *including the majority*. We introduce a new and important definition of fairness which is that a debiasing algorithm should remove biases while *at least not degrading performance for any community*. Performance at the community level is not a fungible commodity.

Overview of our proposed algorithm:

We propose a new debiasing algorithm in the pre-processing category that addresses these limitations and has the following innovations. (1) It operates in a multi-community setting. (2) Minority and majority communities are determined for each classifier: model - dataset combination. When a classifier results in an imbalance in false positive rates (FPR) for a dataset there is bias; communities suffering higher rates are considered as minority. (3) Our algorithm is fair towards all communities because it addresses *all* false positive errors made by the classifier irrespective of source community. In contrast, competing algorithms (Kamiran and Calders, 2012; Ball-Burack et al., 2021) involve complex criteria

to adjust training data that rely on community identity. (4) Our algorithm is inspired by research on text obfuscation algorithms. Specifically, we add obfuscated versions of false positive instances to the training data. This ensures that the synthetic instance is as close as possible to the original while increasing the likelihood of the classifier changing its (false) positive decision to a (true) negative for the instance. As a secondary objective we explore the relative merits of different obfuscators in the context of debiasing classifiers.

Contributions of our research:

1. We propose an obfuscation based debiasing approach designed to handle multiple communities¹. It is fair in that it does not sacrifice performance for the minority or majority communities.
2. We test our ideas through experiments on debiasing (for racial/ethnicity bias) in binary toxicity classifiers built using combinations of 5 models and 3 datasets. We also assess the relative merits of 4 obfuscators used for debiasing. We compare against four state-of-the-art pre-processing baseline approaches: Preferential Sampling (Kamiran and Calders, 2012), Differential Tweetment (Ball-Burack et al., 2021), SMOTE (Chawla et al., 2002), and Counterfactual Data Augmentation.

In summary, results indicate that our community neutral algorithm successfully reduces bias while simultaneously reducing FPR and at least maintaining F1 scores for *majority and minority communities*. Baselines make large sacrifices in performance in order to reduce bias. Regardless of obfuscating classifier, our approach almost always debiases and meets our fairness requirement.

2 Desired properties of debiasing algorithms

2.1 Multi-community perspective

Any given dataset or classifier model may be biased against multiple communities simultaneously and not just one. Thus, a multi-community perspective is crucial. Unfortunately, extensions to existing debiasing algorithms for supporting a multiple community perspective are generally not obvious or straightforward (Mozafari et al., 2020; Ball-Burack et al., 2021; Halevy et al., 2021). Thus, there is a

¹Our code is available at <https://github.com/IngroJShrestha/debiasing-via-obfuscation-in-multi-community-settings>

critical need for algorithms that are designed for multi-community settings.

2.2 Definition of fairness

Our definition of *fairness* requires debiasing algorithms to reduce bias without at least degrading performance for any community. To illustrate, consider the popular strategy of measuring bias by false positive rate (FPR). We can think of several approaches to reduce the FPR for a minority community. But those that involve increasing FPR for other communities (including the majority) or that decrease F1 scores for any community are not appropriate. Moreover, by sacrificing the majority we may also be sacrificing any ‘yet to be identified’ latent minority groups.

Unfortunately, most debiasing papers do not report performance on the majority community. Typically, papers report bias measurements and model performance for the whole dataset or/and the minority subset (Dixon et al., 2018; Park et al., 2018; Savani et al., 2020; Xia et al., 2020; Ball-Burack et al., 2021; Sen et al., 2022; Song et al., 2023; Sobhani and Delany, 2024) with rare exceptions (Mozafari et al., 2020; Halevy et al., 2021; Tang et al., 2024). Thus, we cannot gauge if these are fair as per our definition.

We hope to convince future researchers to consider multiple communities simultaneously and also that the fairness property we outline is the right one to pursue. Our results in the context of debiasing toxicity classifiers show that this is possible.

3 Debiasing Methodology

3.1 Overview of our approach

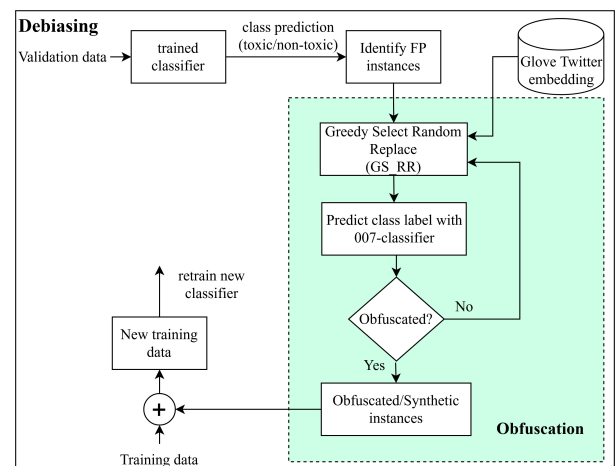


Figure 1: Overall architecture

Our approach is a pre-processing one as we adjust training data. Minority/majority communities are identified by FPR (*Type-I error rate*) as is common in the literature (Sap et al., 2019; Xia et al., 2020; Halevy et al., 2021; Mehrabi et al., 2021; Lwowski et al., 2022; Spliethöver et al., 2024). Communities with higher FPR are considered as minority and the rest as majority (using a suitable threshold described later). Here FP, which is in the context of toxicity classifiers, represents a *non-toxic* text falsely predicted as *toxic*. FPR is used instead of false negative rate (FNR) since the punitive impact of a false positive decision on the individual outweighs the impact of a false negative decision on the organization, especially with toxicity (Davani et al., 2023).

We use methods inspired by obfuscation algorithms to generate synthetic instances. Specifically, each FP instance is modified until a target classifier is tricked into changing its decision. The choice of target classifier for use in obfuscation is independent of the classifier being debiased. We minimize confusion between the two by referring to the target classifier henceforth as 007-classifier (for a bit of fun and since obfuscation is typically adversarial in nature). We explore alternate 007-classifiers (Section 3.1.2).

We harvest FPs from validation data (instead of training). Key to note, the validation datasets provide fewer FPs than the training datasets. For the 3 datasets we investigate, the number of validation set FPs is 1.4 to 3.5 times smaller than the number of training set FPs. This may be a disadvantage for our algorithm. However, it has the potential advantage in that the FP examples from validation data are ‘unseen’ by the model and hence might lead to better generalization for our debiasing objective.

As shown in figure 1, we process the validation dataset with the initial toxicity classifier, add obfuscated versions of all FP instances to the training data and retrain the classifier. We expect this to be less biased without sacrificing performance.

3.1.1 Obfuscation: greedy select random replace

Text obfuscation algorithms have been developed for adversarial attacks against classifiers to force them into making errors. A large body of research explores obfuscation attacks against classifiers of text attributes such as author, gender, and offensiveness (Mahmood et al., 2019; Rusert et al., 2022; Tokpo and Calders, 2022; Xing et al., 2024). Instead

of adversarial attacks we plan to use obfuscators in a novel manner for debiasing classifiers. Specifically, we use obfuscators to generate synthetic instances that are as close to the FP instances as possible while raising the likelihood of the biased classifier flipping its decision. Thus, in contrast to using obfuscators adversarially, our application of obfuscators to debiasing is novel in that it is beneficial.

We follow a greedy-select random-replace obfuscation strategy (see Figure 1) wherein we (1) select an appropriate term in the instance and (2) substitute it with a suitable replacement. If the 007-classifier alters its decision on this modified instance then obfuscation concludes else we iterate back into the greedy-select random-replace step. For step 1, the algorithm greedily selects the one word whose removal has greatest impact on 007-classifier confidence. Step 2 iterates through random selections for substitution from Glove Twitter². Random replacements are efficient since these synthetic instances will never be viewed by humans. The cycle continues until the obfuscation classifier is tricked. The degenerate case of not tricking the 007-classifier never happens in our experiments. Appendix Table 9 presents examples of FPs and their obfuscated version (synthetic instances). We emphasize that the substitutions are not intended to be sensible - as these are random word choices.

3.1.2 Alternative 007-classifiers

A secondary goal of this study is to understand if domain differences between the 007-classifiers and the classifier being debiased matter.

1. The 007-classifier is identical to the classifier being debiased (OBF_{TC}). This reflects a white box assumption wherein we have full access to the classifier being debiased.
2. The 007-classifier is from a related domain. This reflects a black box assumption in which we are given a set of FPs and we only know what kind of classifier is being debiasing. Since this is a toxicity classifier, we explore MIDAS (Mahata et al., 2019) and NULI (Liu et al., 2019a), two state-of-the-art offense classifiers from OffensEval 2019 Task 6 (Zampieri et al., 2019) (OBF_{MIDAS}, OBF_{NULI}). NULI, a BERT based classifier, was the top performing offense classifier. MIDAS, the top performing non-BERT offense classifier, is a BLSTM, BLSTM + BGRU, CNN ensemble.

²<https://nlp.stanford.edu/projects/glove/>

3. The 007-classifier is from a different domain. This is also black box in that we are given FP instances but have no knowledge of the classifier’s problem domain. Thus, we use a popular classifier, i.e., a sentiment classifier as the 007-classifier (OBF_{SC}) (Barbieri et al., 2020).

3.2 Baselines

We use pre-processing bias mitigation methods that adjust the training dataset as our baselines.

Preferential Sampling (PS) (Kamiran and Calders, 2012): They reduce bias by deleting or adding exact copies of select instances. Selections are based on instance rankings by toxic class membership probabilities. As an illustration, a number (calculated value) of top-ranked toxic instances are removed for the minority group if it has more toxic comments compared to the majority group. Details are in Kamiran and Calders (2012).

The authors only consider African-American (AE) as *minority* and *non-AE* as *majority*. Thus, as a simple extension we run experiments where each minority community (which varies by model - dataset combination) in turn forms the minority group and the rest form the other group. These are labeled as PS_{Hispanic}, PS_{AE}, etc. These experiments will alert us to differences in bias reduction from different minority perspectives. The authors also indicate that multiple groups may be combined such that only two groups remain. Accordingly, for each dataset, we combine communities with high false positive rates into one minority group keeping the rest as one majority group (PS_{allMin}).

Differential Tweetment (DiffT) (Ball-Burack et al., 2021): This strategy which derives from PS (Kamiran and Calders, 2012) also deletes or adds exact copies of training instances after ranking. Instance ranking is done based on both classifier confidence and probability of belonging to the minority group. Ranking and selection is iterative, with the top 1000 instances selected for processing (removal or duplication). The classifier is re-trained. If bias is below a threshold, the process terminates. Details of our implementation are in the appendix A.1³.

The authors consider bias measurement only from *AE* perspective. Unlike the PS algorithm, we do not find an intuitive way to extend DiffT wherein multiple minority groups can be combined into one group. Accordingly, we only consider

debiasing from the perspective of each minority group (labeled DiffT_{Asian}, DiffT_{Hispanic} etc.).

SMOTE (Chawla et al., 2002): While SMOTE is designed to handle class imbalance we are curious to see if it can also be used to reduce bias. SMOTE adds new synthetic instances as follows. Each instance in the smaller class is paired with its closest $k \in [1, 5]$ instances from the same class in the feature space. In our case, the classes are *toxic* and *non-toxic*. A synthetic sample is generated — in this feature space — along the line between each pair using the equation:

$$x_{new}^j = x_i^j + \text{rand}(0, 1) * (x_i^j - x_{neighbors}^j)$$

Here x_i^j refers to j -th feature of a sample $x_i \in \mathbb{R}^n$, $j \in [1, n]$. $x_{neighbors}$ refers to one of the k -chosen neighbors, and x_{new} refers to a synthetic sample in the line between x_i and $x_{neighbors}$.

Counterfactual Data Augmentation (CDA):

Aligned with our goal of adding obfuscated FPs, we added the counterfactual of FPs from the validation set to the training dataset. That is, we added the FP text without modification but with the label ‘toxic’.

4 Experiment design

4.1 Toxicity classifier models

In the first set of experiments, we debias a multi-layer perceptron (MLP) classifier and make detailed comparisons with baseline debiasing algorithms (Section 5). In the second set of experiments, we extend our analysis to debiasing masked language models (MLMs) classifiers: BERT (bert-base-uncased, bert-large-uncased) (Devlin et al., 2019), DistilBERT (Sanh et al., 2019), RoBERTa (roberta-base) (Liu et al., 2019b) (Section 6). Model configurations are in the appendix A.2.

4.2 Toxicity classifier datasets

We experiment with three datasets. As in Park et al. (2018), Senarath and Purohit (2020) and Fortuna et al. (2021), we formulate toxicity classification as binary (*toxic/non-toxic*) instead of multi-class and adjust the class definition accordingly in each dataset. A *toxic* text contains hateful, abusive, or offensive content⁴. Table 1 presents an overview of datasets used. Each was randomly split into training (65%), validation (15%) and testing (20%).

³We held discussions over email with the authors to clarify specific steps in their algorithm.

⁴For example, *Are you f**king stupid, dude?* is a toxic text as it contains offensive language and demeaning phrases that can be perceived as abusive towards an individual.

(1) **Davidson (DWMW17)** (Davidson et al., 2017): It contains approximately 25K tweets each annotated by at least three CrowdFlower (CF) workers. Tweets are labeled as one of *hate speech*, *offensive* or *none*. We combined *hate speech* and *offensive* as *toxic* and the rest as *non-toxic*.

(2) **HatEval (HatEval19)** (Basile et al., 2019): It has about 13K English tweets with two labels: *offensive* or *not* annotated using crowdsourcing platform *Figure Eight* and two more experts. We consider *offensive* as *toxic* and *not* as *non-toxic*.

(3) **Founta (FDCL18)** (Founta et al., 2018): It includes approximately 100K tweets, each annotated by five crowdsource workers as *hateful*, *abusive*, *spam* and *none*. We combine *hateful* and *abusive* as *toxic* and *spam* and *none* as *non-toxic*.

Dataset	Class	Group	Train	Valid	Test	Total
DWMW17	toxic	AE	8178	1887	2463	12528
		Hispanic	2128	456	693	3277
		Asian	106	33	36	175
		White	2946	707	918	4571
		Total	13358	3083	4110	20551
	non-toxic	AE	425	97	125	647
		Hispanic	241	59	71	371
		Asian	152	25	47	224
		White	1868	439	584	2891
		Total	2286	620	827	4144
HatEval19	toxic	AE	714	171	215	1100
		Hispanic	443	103	129	675
		Asian	60	14	14	88
		White	2328	531	733	3592
		Total	3545	819	1091	5455
	non-toxic	AE	477	117	125	719
		Hispanic	485	109	161	755
		Asian	219	55	74	348
		White	3681	841	1136	5658
		Total	4862	1122	1496	7480
FDCL18	toxic	AE	5275	1206	1653	8134
		Hispanic	5807	1373	1756	8936
		Asian	384	61	110	555
		White	9401	2175	2902	14478
		Total	20867	4815	6421	32103
	non-toxic	AE	2032	455	655	3142
		Hispanic	2416	563	734	3714
		Asian	4188	978	1375	6541
		White	35110	8100	10696	53906
		Total	41714	10096	12805	67303

Table 1: Description of datasets.

Pre-processing: We replaced new line and extra white space with a space. We removed URLs, hashes, mentions (@user), and retweet mention (RT) from the tweet, then lower case and stem text.

Dialect estimation: We use the model from Blodgett et al. (2016) to infer dialect⁵ on the pre-

⁵Blodgett et al. (2016) found strong correlations between predicted and the actual census proportions for AE, Hispanic, and White, and borderline weak to moderate for Asian, suggesting our Asian texts may be blurred with the background. Further linguistic study is needed to improve the estimation of

processed tweets. We take *argmax* of the probability distribution that it provides for: African American (AE; p_{AE}), Hispanic(H; $p_{Hispanic}$), Asian(AS; p_{Asian}), and White (p_{White}) for group assignment.

Majority and minority categorization: Generally, papers use a static, socially recognized demarcation with disadvantaged group African Americans as the minority and advantaged Whites as the majority. Hispanics and Asians are generally less studied. In contrast, we use classifier performance to assign minority and majority groups. This assignment is model - dataset specific based on the model’s FPR values in the validation subset. Communities with relatively higher FPR are regarded as *minority* and the others *majority*. Using 50% of the largest FPR value as threshold (t), communities with $FPR \geq t$ are regarded as minority and the rest as majority. With very similar FPR values, e.g., 0.40, 0.35, 0.29, 0.27 then we could use a more stringent threshold as for example 75% of highest FPR; this assigns the top 2 FPR scores to minority communities and the rest majority. For us 50% threshold separates communities into two groups.

4.3 Performance measurement

Let $S = \{C_1, C_2, \dots, C_n\}$ be the set of communities of interest. In our case, $n = 4$ (AE, Hispanic (H), Asian (AS), White). We use the commonly accepted notion of community specified by race/ethnicity. We measure average FPR across majority and across minority communities and across all communities. Bias (B) is given by $FPR_{minority} - FPR_{majority}$. We note that bias can be close to 0 with both $FPR_{majority}$ and $FPR_{minority}$ being high (undesired) or low (desired). We also gauge performance with macro F1 scores computing averages $F1_{all}$, $F1_{majority}$ and $F1_{minority}$.

4.4 Viable debiasing run

Formally, we define a viable debiasing algorithm as one that reduces bias by at least 4%⁶. Additionally, FPRs ($FPR_{majority}$, $FPR_{minority}$ and FPR_{all}) should not increase. Lastly, F1 performance should not decline. Since this is likely very challenging, we adopt a soft criteria, considering a strategy viable if

Asian dialect. Key to note here is that our approach effectively mitigates bias for this ‘group’, even if its Asian label is noisy.

⁶We looked to the literature to set a threshold. The lowest FPR based bias reduction declared significant and successful was 3.3% (Chuang et al., 2021; Liu et al., 2021). We round up this value and consider 4% as the minimum threshold to declare significant bias reduction.

the drops in $F1_{\text{majority}}$, $F1_{\text{minority}}$ and $F1_{\text{all}}$ are each at most 5%. Larger drops are too high a sacrifice.

5 Results for debiasing MLP

Tables 2 to 4 present test set results as averages over ten runs to accommodate any variations due to the random initialization of the neural networks. Minority and majority communities identified are specified in table legends. Bias is dataset specific, e.g., there is bias against Asian in DWMW17 but not in the other two datasets. Of note, AE faces the highest bias in all 3 datasets. The number of baseline runs is determined by the number of minority communities and hence these vary across datasets. The "Original" row is for the original (biased) model. *Sacrifices* records penalties paid.

Impact on bias: Across datasets, most of the 23 baseline runs reduce bias by 4% or more (indicated by ●). The 7 exceptions are marked with ○: e.g., PS_H in all datasets. More than half of the successful runs reduce bias extremely well, by at least 10%, e.g., PS_{AE} , PS_{allMin} , and SMOTE across all datasets.

In contrast, *all* of our obfuscation based debiasing strategies reduce bias. These range from 6.3% - 11.3% (DWMW17), 8.6%-11.9% (FDCL18), and 11.9%-13.7% (HatEval19). More than half of these reduce bias extremely well, i.e., by at least 10%. The lowest was achieved by $OB_{\text{NULI}}/OB_{\text{TC}}$ strategy for DWMW17 and the highest by OB_{TC} for HatEval19. Considering bias reduction alone, 16/23 baseline runs and all of our 12 runs are *potentially viable*. We limit further analysis to these runs.

Impact on FPR: Sacrifices in FPR are marked by red diamonds (◇: < 5% increase in FPR, ◇: [5,10)% increase, ◆: ≥ 10% increase) ordered by sacrifices in FPR_{majority} , FPR_{minority} , FPR_{all} .

For DWMW17, all three FPRs become worse for all baselines except SMOTE. In FDCL18 and HatEval19, except for DiffT, all baselines incur at least one FPR penalty. Across datasets, 19/27 negative impacts on FPR are high. For example, PS_{AE} and PS_{allMin} in DWMW17 reduce bias extremely well but with huge FPR penalty (e.g., FPR_{majority} penalty is about 223% for PS_{AE}). Likewise, SMOTE's reduction of bias by 43.4% in FDCL18 pays a huge FPR penalty⁷ of more than 200%.

In contrast, all our debiasing strategies reduce

⁷In preliminary experiments we found that Fair-SMOTE (Chakraborty et al., 2021) also was not viable because of performance drops. Moreover, their Tables 5-6 show sacrifices in accuracy and FPR while reducing bias.

all FPRs. Approximately 50% of the drops are at least 10%. All but one of the remaining drops are in [5%, 10%). For example, OB_{TC} in FDCL18 drops FPR_{majority} by 19%, FPR_{minority} by 10.4% and FPR_{all} by 11%. Average of all three FPRs reductions across 007-classifiers is largest for FDCL18 (14%) and (7%-8%) for the other two datasets.

In sum, 5 of the 17 baseline runs that reduce bias stay potentially viable after considering FPR changes. In contrast all of our 12 runs remain potentially viable. We limit further analysis to these runs.

Approach	Majority		Minority		All		Bias (B)	Sacrifices		
	FPR	F1	FPR	F1	FPR	F1		B	F1	FPR
Original	0.095	0.887	0.207	0.876	0.179	0.879	0.112			
PS_{AE}	+223	-7.7	+70.0	-7.4	+90.5	-7.5	-59.8	●	xxx	◆◆◆
PS_H	+24.2	-0.2	+24.6	-3.0	+24.6	-2.3	+25.0	○		◆◆◆
PS_{AS}	+6.3	0.0	+1.0	+0.3	+1.7	+0.2	-3.6	○		◇◇◇
PS_{allMin}	+178	-8.6	+7.2	-4.3	+29.6	-4.7	-138	●	x	◆◇◆
DiffT _{AE}	+28.4	-0.5	+9.2	-1.3	+11.7	-1.1	-7.1	●		◆◇◆
DiffT _H	+24.2	-0.3	+2.9	-0.7	+5.6	+0.1	-15.2	●		◆◇◇
DiffT _{AS}	+4.2	0.0	+3.4	-0.7	+3.9	-0.6	+2.7	○		◇◇◇
SMOTE	-8.4	-21.5	-39.6	-21.2	-35.2	-21.4	-66.1	●	xxx	
CDA	+16.8	-0.3	+2.9	+0.2	+5.0	0.0	-8.9	●		◆◇◇
OB_{TC}	-5.3	+0.2	-5.8	0.0	-5.6	0.0	-6.3	●		
OB_{MIDAS}	-9.5	+0.2	-10.5	+0.2	-10.4	+0.2	-11.3	●		
OB_{NULI}	-10.8	+0.3	-8.4	+0.2	-8.7	+0.3	-6.3	●		
OB_{SC}	-8.4	+0.3	-7.7	+0.2	-7.8	+0.2	-7.1	●		

Table 2: **DWMW17** test set results. Minority (Original FPR): AE (0.239), Hispanic (H) (0.217), Asian (AS) (0.166), Majority (Original FPR): White (0.095). Cell values indicate percentage increase or decrease from *Original* for all metrics. ○: bias reduction below 4%, ●: bias reduction ≥ 4%, x: F1 drop > 5%, ◇: FPR penalty below 5%, ◇: FPR penalty in the range [5%,10%), ◆: FPR penalty ≥ 10%. Penalties are ordered as Majority, Minority then All. Bias reduction results in **bold** represent viable debiasing runs.

Approach	Majority		Minority		All		Bias (B)	Sacrifices		
	FPR	F1	FPR	F1	FPR	F1		B	F1	FPR
Original	0.021	0.904	0.164	0.901	0.092	0.901	0.143			
PS_{AE}	+81.0	-1.7	-0.6	-7.2	+8.7	-4.4	-12.6	●	x	◆◆◆
PS_H	+90.5	-2.5	+14.6	-5.3	+56.5	-3.9	+3.5	○	x	◆◆◆
PS_{allMin}	+152	-4.3	-24.4	-12.7	-3.3	-8.4	-50.3	●	xx	◆
DiffT _{AE}	-52.4	-2.7	-45.1	-11.2	-45.7	-6.9	-44.1	●	xx	
DiffT _H	-61.9	-9.7	-56.1	-15.1	-56.5	-12.4	-54.5	●	xxx	
SMOTE	+2329	-43.9	+260	-19.4	+499	-31.7	-43.4	●	xxx	◆◆◆
CDA	+52.4	-1.0	+23.2	-0.6	+27.2	-0.8	+18.2	○		◆◆◆
OB_{TC}	-19.0	-0.7	-10.4	-0.6	-10.9	-0.6	-9.1	●		
OB_{MIDAS}	-19.0	-0.9	-12.2	-0.4	-12.6	-0.6	-11.2	●		
OB_{NULI}	-18.1	-0.9	-9.8	-0.5	-10.3	-0.6	-8.6	●		
OB_{SC}	-19.0	-0.9	-12.8	-0.4	-13.0	-0.7	-11.9	●		

Table 3: **FDCL18** test set results. Minority (Original FPR): AE (0.180), Hispanic (0.147), Majority (Original FPR): White (0.028), Asian (0.013). Please refer to Table 2 for explanation of cell values and notation in *Sacrifices* column.

Approach	Majority		Minority		All		Bias (B)	Sacrifices		
	FPR	F1	FPR	F1	FPR	F1		B	F1	FPR
Original	0.133	0.583	0.322	0.652	0.227	0.617	0.190			
PS _{AE}	+19.5	+3.6	-4.0	-4.1	+3.1	-0.5	-21.1	●	♦	♦
PS _H	+3.8	-0.2	+3.1	+0.9	+3.5	+0.5	+2.1	○	♦	♦
PS _{allMin}	+15.0	+10.1	-15.5	-4.3	-6.6	+2.6	-36.8	●	♦	
DiffT _{AE}	-91.0	-15.4	-93.5	-37.9	-93.0	-27.2	-95.3	●	xxx	
DiffT _H	-9.0	+0.3	-5.9	+0.3	-6.6	+0.3	-4.2	●		
SMOTE	+130	-3.3	+45.7	-1.1	+70.9	-2.1	-14.2	●	♦	♦
CDA	+32.3	+7.7	+15.8	-3.1	+21.1	2+1	+4.2	○	♦	♦
OBFT _C	-7.5	+3.8	-10.6	+0.3	-9.3	+2.1	-13.7	●		
OBFM _{IDAS}	-2.0	+5.5	-7.7	-0.6	-5.9	+2.4	-12.2	●		
OBFN _{ULI}	-4.6	+3.2	-8.6	-0.5	-7.2	+1.3	-11.9	●		
OBFS _C	-4.5	+5.5	-9.0	-0.3	-7.5	+2.4	-12.6	●		

Table 4: **HatEval19** test set results. Minority (Original FPR): AE (0.389), Hispanic (0.255), Majority (Original FPR): Asian (0.093), White (0.172). Please refer to Table 2 for explanation of cell values and notation of *Sacrifices* column.

Impact on F1: Runs decreasing F1 score by 5% or more are depicted by $\times$ under Sacrifices in the order of $F1_{\text{majority}}$, $F1_{\text{minority}}$ and $F1_{\text{all}}$.

All but one of the remaining 5 baselines pay large F1 penalties. E.g., SMOTE reduces all three F1s by approximately 21% in DWMW17, while DiffT_{AE} reduce them by 15.4%-38% in HatEval19. The only baseline run that remains viable when considering bias reduction and losses in FPR and F1 is DiffT_H - limited to the HatEval19 dataset.

In contrast, all of our 12 obfuscation runs remain viable even after considering F1 performance, with drops of at most 0.9% across all three F1s. A small point: F1 even improves in HatEval19; by 3.2% - 5.5% ($F1_{\text{majority}}$) and by 1.3% - 2.4% ($F1_{\text{all}}$).

5.1 Sensitivity Analysis

We now explore the effect of processing fewer than *all* FP instances (our base strategy); we limit analysis to the run with MLP as 007-classifier. We explore bias reduction with random FP samples of size (s); start at 20 and increase in steps of 20 with 10 repetitions for each s to account for sample variations. Figures in appendix A.3 show a gradual decrease in bias as more FP instances are processed. This supports our approach of obfuscating all FPs.

Summary for MLP debiasing results: Most baseline runs reduce bias effectively, but sacrifice F1 and/or FPR hindering their overall success. The one exception: DiffT_H for HatEval19 offers a small reduction of bias (4.2%). Key to note, performance varies depending on minority perspective (e.g., see variations in the tables across the three single community perspectives with PS for DWMW17). Debiasing with an AE perspective, dominant in prior

studies, sacrifices heavily in our multi-community and fairness context. In contrast to baselines, all of our obfuscation-based approaches are viable, reducing bias by 6.3%-13.7% without compromising F1 or FPR.

6 Results for debiasing MLMs

We now extend our research to debiasing MLMs of varying size (results in Appendix Tables 6 to 8). Unlike with MLP (5-18M parameters) where all of our runs remain viable, with MLMs a few are not. Also for all MLM - dataset combinations the minority communities are AE and Hispanic and the majority White and Asian (see Table 5).

Bias reduces the most for DistilBERT (66M) (4%-13.8%) and all 12 runs are viable. The next most effective bias reduction is for BERT-base (110M), followed by RoBERTa-base (125M) - for both bias reduces by 4%-13.3%. The 3 to 4 exceptions are due to not meeting the bias reduction threshold or FPR penalty. For example, OBFT_C in RoBERTa-base and OBF_{MIDAS} in BERT-base penalize FPR_{majority} by 4%-5%. Relatively speaking, bias mitigation is least successful in BERT-large (340M), the largest MLM analyzed. But even here, close to 60% of the 12 runs are viable with bias reduction in 4%-11.4%. As examples of exceptions, OBFS_C (FDCL18 and HatEval19) and OBF_{NULI} (FDCL18) do not reduce bias sufficiently while in DWMW17 a couple of runs incur FPR penalties. Possibly, relative to its size, the number of FPs available for obfuscation with BERT-large is insufficient. However, performance in the last three MLMs varies only slightly.

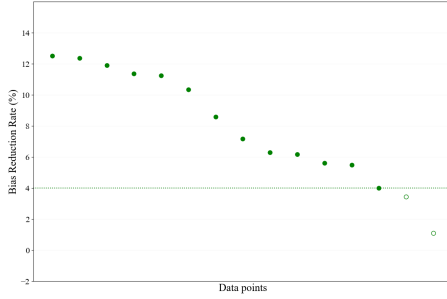
Summary for MLP and MLM results: All our runs are viable with MLP and DistilBERT. As model size increases, bias mitigation is slightly challenging. Interestingly, our debiasing strategy never incurs F1 penalties. However, there are a few instances where our strategies incur FPR penalties.

7 Additional analysis

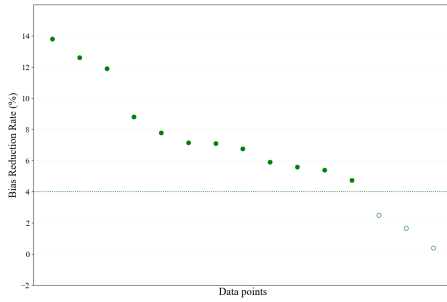
7.1 Performance by 007-classifiers

Figure 2 shows that NULI as the 007-classifier is the best, succeeding in all but two runs. With each remaining 007-classifier, including one built from the same model that is being debiased (OBFT_C) and the sentiment classifier (OBFS_C), approximately 80% of runs are viable. The few failures are due to insufficient bias reduction or/and FPR losses. Interestingly, while OBF_{NULI} and OBF_{MIDAS} are

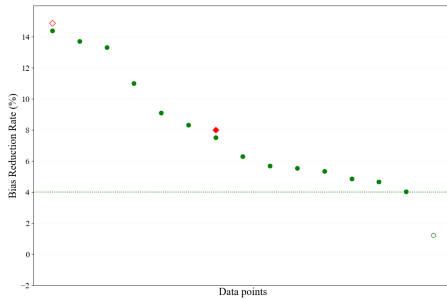
both offense classifiers trained on the same data, performance varies – possibly due to architectural differences since MIDAS is an older non BERT ensemble. Considering all viable runs (48/60) across 007-classifiers, average bias reduction is 8.2% (standard deviation, SD: 3.1). Average minimum bias reduction is 4.2% (SD: 0.31), and average maximum is 13.1% (SD: 0.70). The choice of 007-classifier does not make a big difference.



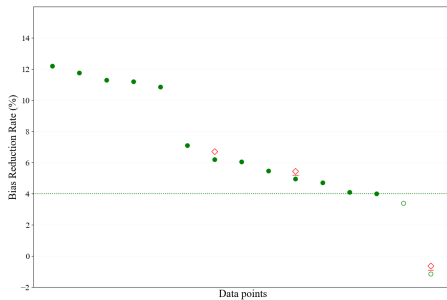
(a) Using NULI as 007-classifier (OBF_{NULI})



(b) Using Sentiment Classifier as 007-classifier (OBF_{SC})

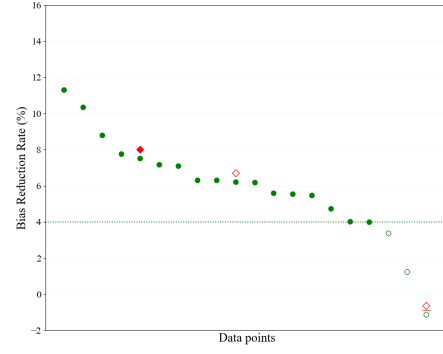


(c) Using model as 007-classifier (OBF_{TC})

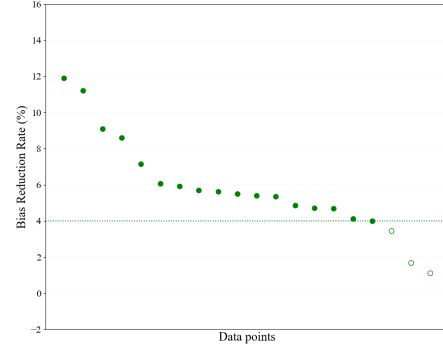


(d) Using MIDAS as 007-classifier (OBF_{MIDAS})

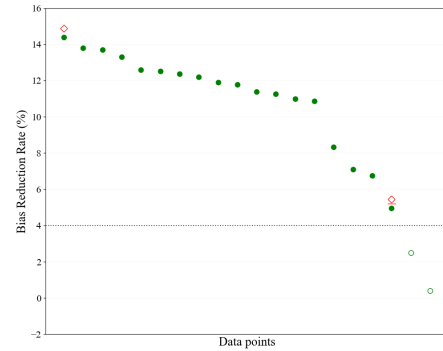
Figure 2: Performance by 007-classifiers. Please refer to Table 2 for explanation of notations.



(a) DWMW17



(b) FDCL18



(c) HatEval19

Figure 3: Performance by dataset. Please refer to Table 2 for explanation of notations.

7.2 Performance by dataset

As shown in Figure 3, FDCL18 yields the best performance; bias reduces for all but 3 of the 20 runs with no penalties. In DWMW17 and HatEval19, most runs are viable. Exceptions are due to insufficient bias reduction or/and sacrifice in FPR.

Average bias reduction across successful runs is highest in HatEval19 (11.3%; SD: 2.08) and around 6.6% (SD: 2.2) for the other two. Not surprisingly, HatEval19 is the most biased initially (e.g., BERT-large has an initial bias value of 0.26 while it is 0.12 and 0.15 for DWMW17 and FDCL18, respectively).

8 Related works

We briefly review pre-processing methods or debiasing text classifiers and their two desired properties.

Pre-processing debiasing The most common strategy is *replicate/remove instances*, which includes but is not limited to PS (Kamiran and Calders, 2012), DiffT (Ball-Burack et al., 2021), Fair-SMOTE (Chakraborty et al., 2021) and Counterfactual Data Augmentation (Xie et al., 2023; Sobhani and Delany, 2024). Due to their importance in NLP *remove stereotypical bias in word embeddings* (Bolukbasi et al., 2016; Brunet et al., 2019) is another common approach. Other less common approaches include *relabel instances* (Kamiran and Calders, 2009; Luong et al., 2011), *re-weighting* (Kamiran and Calders, 2012; Almuzaini et al., 2022), and *shift distribution* of unprotected attributes such that sensitive attributes cannot be estimated (Feldman et al., 2015). Our baselines are state-of-art strategies closest to ours in spirit.

Multi-community perspective: Debiasing papers focus exclusively on the AE minority community (Xia et al., 2020; Ball-Burack et al., 2021; Halevy et al., 2021; Spliethöver et al., 2024). Extension of algorithms to the multi-community setting is not obvious. E.g., Ball-Burack et al. (2021) rank instances (before selecting ones for deletion/replication) on the basis of a single community (p_{AE}). It is not clear how to extend this ranking criteria for multiple minority communities. In contrast, we are able to make a reasonable extension to PS (Kamiran and Calders, 2012) although it did not fare well. Overall, the debiasing literature has not paid attention to this important property.

Fairness to all communities: Most debiasing papers do not report measurements for *both* the majority and minority groups (Kamiran and Calders, 2012; Calmon et al., 2017; Park et al., 2018; Dixon et al., 2018; Savani et al., 2020; Shrestha et al., 2022; Ball-Burack et al., 2021; Cheng et al., 2022; Song et al., 2023; Sobhani and Delany, 2024; Iskander et al., 2024). Thus, we do not know whether these satisfy our fairness property. Overall, it is clear from the literature that our perspective on fairness in performance has not been a concern.

9 Conclusion

We show that it is possible to debias toxicity classifiers in a multi-community setting without sacrificing performance for either minority or majority

communities. We achieve this with a novel debiasing algorithm involving text obfuscators. Key to note, we use obfuscators in a beneficial manner in contrast to prior use of obfuscators to adversarially attack classifiers. Our experiments with toxicity classifiers built using combinations of 5 neural models and 3 datasets and debiased using 4 obfuscators while considering 4 communities, yield excellent results; the few failures occur when debiasing the larger MLMs. In contrast, state-of-the-art baselines and their variants perform poorly; these reduce bias nicely but make large sacrifices in FPR and F1. Overall, we conclude that it is possible to debias via obfuscation while being fair to both minority and majority communities. While we have shown that it is possible to debias toxicity classifiers in the context of race/ethnicity bias, debiasing these classifiers for other dimensions, such as gender, is left to future research.

10 Limitations

Our work is focused on a binary classification task: toxic or not. Extending it to a multi-class setting and exploring other domains is left for future work. We also address bias only for race/ethnicity. Further exploration is needed for other sensitive attributes, such as gender and religion.

Another limitation is that, instead of the validation set as the source, we could use other similar toxicity data to generate synthetic, non-toxic instances. We leave this ‘distance learning’ inspired extension to future work.

Additionally, we do not explore our bias mitigation approach in the context of large generative language models like GPT-4 (Achiam et al., 2023), Llama 3 (Dubey et al., 2024), and Mistral (Jiang et al., 2023). These have different challenges due to their very large architectures and their vast training corpus which are often underspecified. Hence these are likely to require very different debiasing strategies. We leave this exploration for the future.

11 Ethical consideration

Our work focuses on mitigating race/ethnicity bias across four communities assessed in the context of toxicity classification. We aim to address biases faced by minority communities while also being fair to the majority groups. Our research raises the likelihood of classifiers operating more ethically.

While our approach successfully mitigates race/ethnicity bias in the toxicity classifiers, we

acknowledge that the classifiers may still exhibit bias in other dimensions, such as gender and age, or in intersectional dimensions (e.g., gender combined with race/ethnicity). Our method specifically address racial bias in MLP/MLM toxicity classifiers; therefore, claims of bias mitigation should not be interpreted as the complete removal of all biases. This underscores the necessity of addressing all biases comprehensively before deploying these classifiers.

References

- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. GPT-4 Technical Report. *arXiv preprint arXiv:2303.08774*.
- Abdulaziz A Almuzaini, Chidansh A Bhatt, David M Pennock, and Vivek K Singh. 2022. Abcinml: Anticipatory bias correction in machine learning applications. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, pages 1552–1560.
- Ari Ball-Burack, Michelle Seng Ah Lee, Jennifer Cobbe, and Jatinder Singh. 2021. Differential tweetment: Mitigating racial dialect bias in harmful tweet detection. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pages 116–128.
- Francesco Barbieri, Jose Camacho-Collados, Luis Espinosa Anke, and Leonardo Neves. 2020. [TweetEval: Unified benchmark and comparative evaluation for tweet classification](#). In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1644–1650, Online. Association for Computational Linguistics.
- Valerio Basile, Cristina Bosco, Elisabetta Fersini, Debora Nozza, Viviana Patti, Francisco Manuel Rangel Pardo, Paolo Rosso, and Manuela Sanguinetti. 2019. [SemEval-2019 task 5: Multilingual detection of hate speech against immigrants and women in Twitter](#). In *Proceedings of the 13th International Workshop on Semantic Evaluation*, pages 54–63, Minneapolis, Minnesota, USA. Association for Computational Linguistics.
- Su Lin Blodgett, Lisa Green, and Brendan O’Connor. 2016. [Demographic dialectal variation in social media: A case study of African-American English](#). In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 1119–1130, Austin, Texas. Association for Computational Linguistics.
- Tolga Bolukbasi, Kai-Wei Chang, James Y Zou, Venkatesh Saligrama, and Adam T Kalai. 2016. Man is to computer programmer as woman is to homemaker? debiasing word embeddings. *Advances in neural information processing systems*, 29.
- Marc-Etienne Brunet, Colleen Alkalay-Houlihan, Ashton Anderson, and Richard Zemel. 2019. Understanding the origins of bias in word embeddings. In *International Conference on Machine Learning*, pages 803–811. PMLR.
- Flavio Calmon, Dennis Wei, Bhanukiran Vinzamuri, Karthikeyan Natesan Ramamurthy, and Kush R Varshney. 2017. Optimized pre-processing for discrimination prevention. *Advances in neural information processing systems*, 30.
- Joydallya Chakraborty, Suvodeep Majumder, and Tim Menzies. 2021. Bias in machine learning software: why? how? what to do? In *Proceedings of the 29th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, pages 429–440.
- Nitesh V Chawla, Kevin W Bowyer, Lawrence O Hall, and W Philip Kegelmeyer. 2002. Smote: synthetic minority over-sampling technique. *Journal of artificial intelligence research*, 16:321–357.
- Lu Cheng, Suyu Ge, and Huan Liu. 2022. Toward understanding bias correlations for mitigation in nlp. *arXiv preprint arXiv:2205.12391*.
- Yung-Sung Chuang, Mingye Gao, Hongyin Luo, James Glass, Hung-yi Lee, Yun-Nung Chen, and Shang-Wen Li. 2021. [Mitigating biases in toxic language detection through invariant rationalization](#). In *Proceedings of the 5th Workshop on Online Abuse and Harms (WOAH 2021)*, pages 114–120, Online. Association for Computational Linguistics.
- Aida Mostafazadeh Davani, Mohammad Atari, Brendan Kennedy, and Morteza Dehghani. 2023. [Hate speech classifiers learn normative social stereotypes](#). *Transactions of the Association for Computational Linguistics*, 11:300–319.
- Thomas Davidson, Dana Warmusley, Michael Macy, and Ingmar Weber. 2017. Automated hate speech detection and the problem of offensive language. In *Proceedings of the 11th International AAAI Conference on Web and Social Media, ICWSM ’17*, pages 512–515.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [BERT: Pre-training of deep bidirectional transformers for language understanding](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.

- Lucas Dixon, John Li, Jeffrey Sorensen, Nithum Thain, and Lucy Vasserman. 2018. Measuring and mitigating unintended bias in text classification. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society*, pages 67–73.
- Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. 2024. The LLama 3 Herd of Models. *arXiv preprint arXiv:2407.21783*.
- Michael Feldman, Sorelle A Friedler, John Moeller, Carlos Scheidegger, and Suresh Venkatasubramanian. 2015. Certifying and removing disparate impact. In *proceedings of the 21th ACM SIGKDD international conference on knowledge discovery and data mining*, pages 259–268.
- Paula Fortuna, Juan Soler-Company, and Leo Wanner. 2021. How well do hate speech, toxicity, abusive and offensive language classification models generalize across datasets? *Information Processing & Management*, 58(3):102524.
- Antigoni-Maria Founta, Constantinos Djouvas, Despoina Chatzakou, Ilias Leontiadis, Jeremy Blackburn, Gianluca Stringhini, Athena Vakali, Michael Sirivianos, and Nicolas Kourtellis. 2018. Large scale crowdsourcing and characterization of twitter abusive behavior. *ICWSM*.
- Matan Halevy, Camille Harris, Amy Bruckman, Diyi Yang, and Ayanna Howard. 2021. Mitigating racial biases in toxic language detection with an equity-based ensemble framework. *Equity and Access in Algorithms, Mechanisms, and Optimization*.
- Moritz Hardt, Eric Price, and Nati Srebro. 2016. Equality of opportunity in supervised learning. *Advances in neural information processing systems*, 29:3315–3323.
- Shadi Iskander, Kira Radinsky, and Yonatan Belinkov. 2024. [Leveraging prototypical representations for mitigating social bias without demographic information](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*, pages 379–390, Mexico City, Mexico. Association for Computational Linguistics.
- AQ Jiang, A Sablayrolles, A Mensch, C Bamford, DS Chaplot, D de las Casas, F Bressand, G Lengyel, G Lample, L Saulnier, et al. 2023. Mistral 7B. *arXiv preprint arXiv:2310.06825*.
- Faisal Kamiran and Toon Calders. 2009. Classifying without discriminating. In *2009 2nd international conference on computer, control and communication*, pages 1–6. IEEE.
- Faisal Kamiran and Toon Calders. 2012. Data pre-processing techniques for classification without discrimination. *Knowledge and Information Systems*, 33(1):1–33.
- Deepak Kumar, Oleg Lesota, George Zerveas, Daniel Cohen, Carsten Eickhoff, Markus Schedl, and Navid Rekabsaz. 2023. [Parameter-efficient modularised bias mitigation via AdapterFusion](#). In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2738–2751, Dubrovnik, Croatia. Association for Computational Linguistics.
- Haochen Liu, Wei Jin, Hamid Karimi, Zitao Liu, and Jiliang Tang. 2021. [The authors matter: Understanding and mitigating implicit bias in deep text classification](#). In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 74–85, Online. Association for Computational Linguistics.
- Ping Liu, Wen Li, and Liang Zou. 2019a. [NULI at SemEval-2019 task 6: Transfer learning for offensive language detection using bidirectional transformers](#). In *Proceedings of the 13th International Workshop on Semantic Evaluation*, pages 87–91, Minneapolis, Minnesota, USA. Association for Computational Linguistics.
- Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019b. [RoBERTa: A Robustly Optimized BERT Pretraining Approach](#). *CoRR*, abs/1907.11692.
- Pranay K Lohia, Karthikeyan Natesan Ramamurthy, Manish Bhide, Diptikalyan Saha, Kush R Varshney, and Ruchir Puri. 2019. Bias mitigation post-processing for individual and group fairness. In *Icassp 2019-2019 IEEE international conference on acoustics, speech and signal processing (icassp)*, pages 2847–2851. IEEE.
- Binh Thanh Luong, Salvatore Ruggieri, and Franco Turini. 2011. k-NN as an implementation of situation testing for discrimination discovery and prevention. In *Proceedings of the 17th ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 502–510.
- Brandon Lwowski, Paul Rad, and Anthony Rios. 2022. [Measuring geographic performance disparities of offensive language classifiers](#). In *Proceedings of the 29th International Conference on Computational Linguistics*, pages 6600–6616, Gyeongju, Republic of Korea. International Committee on Computational Linguistics.
- Debanjan Mahata, Haimin Zhang, Karan Uppal, Yaman Kumar, Rajiv Ratn Shah, Simra Shahid, Laiba Mehnaz, and Sarthak Anand. 2019. [MIDAS at SemEval-2019 task 6: Identifying offensive posts and targeted offense from Twitter](#). In *Proceedings of the 13th International Workshop on Semantic Evaluation*, pages 683–690, Minneapolis, Minnesota, USA. Association for Computational Linguistics.
- Asad Mahmood, Faizan Ahmad, Zubair Shafiq, Padmini Srinivasan, and Fareed Zaffar. 2019. A girl has no name: Automated authorship obfuscation

- using mutant-x. *Proc. Priv. Enhancing Technol.*, 2019(4):54–71.
- Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. 2021. A survey on bias and fairness in machine learning. *ACM Computing Surveys (CSUR)*, 54(6):1–35.
- Marzieh Mozafari, Reza Farahbakhsh, and Noël Crespi. 2020. Hate speech detection and racial bias mitigation in social media based on bert model. *PloS one*, 15(8):e0237861.
- Ji Ho Park, Jamin Shin, and Pascale Fung. 2018. **Reducing gender bias in abusive language detection**. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2799–2804, Brussels, Belgium. Association for Computational Linguistics.
- Chen Qian, Fuli Feng, Lijie Wen, Chunping Ma, and Pengjun Xie. 2021. **Counterfactual inference for text classification debiasing**. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 5434–5445, Online. Association for Computational Linguistics.
- T. C. Rajapakse. 2019. Simple transformers. <https://github.com/ThilinaRajapakse/simpletransformers>.
- Jonathan Rusert, Zubair Shafiq, and Padmini Srinivasan. 2022. **On the robustness of offensive language classifiers**. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7424–7438, Dublin, Ireland. Association for Computational Linguistics.
- Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *ArXiv*, abs/1910.01108.
- Maarten Sap, Dallas Card, Saadia Gabriel, Yejin Choi, and Noah A. Smith. 2019. **The risk of racial bias in hate speech detection**. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 1668–1678, Florence, Italy. Association for Computational Linguistics.
- Yash Savani, Colin White, and Naveen Sundar Govindarajulu. 2020. **Intra-processing methods for debiasing neural networks**. In *Advances in Neural Information Processing Systems*, volume 33, pages 2798–2810. Curran Associates, Inc.
- Indira Sen, Mattia Samory, Claudia Wagner, and Isabelle Augenstein. 2022. **Counterfactually augmented data and unintended bias: The case of sexism and hate speech detection**. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4716–4726, Seattle, United States. Association for Computational Linguistics.
- Yasas Senarath and Hemant Purohit. 2020. Evaluating semantic feature representations to efficiently detect hate intent on social media. In *2020 IEEE 14th International Conference on Semantic Computing (ICSC)*, pages 199–202. IEEE.
- Robik Shrestha, Kushal Kafle, and Christopher Kanan. 2022. An investigation of critical issues in bias mitigation techniques. *2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 2512–2523.
- Nasim Sobhani and Sarah Delany. 2024. **Towards fairer NLP models: Handling gender bias in classification tasks**. In *Proceedings of the 5th Workshop on Gender Bias in Natural Language Processing (GeBNLP)*, pages 167–178, Bangkok, Thailand. Association for Computational Linguistics.
- Rui Song, Fausto Giunchiglia, Yingji Li, Lida Shi, and Hao Xu. 2023. Measuring and mitigating language model biases in abusive language detection. *Information Processing & Management*, 60(3):103277.
- Maximilian Spliethöver, Sai Nikhil Menon, and Henning Wachsmuth. 2024. **Disentangling dialect from social bias via multitask learning to improve fairness**. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 9294–9313, Bangkok, Thailand and virtual meeting. Association for Computational Linguistics.
- Kunsheng Tang, Wenbo Zhou, Jie Zhang, Aishan Liu, Gelei Deng, Shuai Li, Peigui Qi, Weiming Zhang, Tianwei Zhang, and Nenghai Yu. 2024. Gendercare: A comprehensive framework for assessing and reducing gender bias in large language models. *arXiv preprint arXiv:2408.12494*.
- Ewoenam Kwaku Tokpo and Toon Calders. 2022. **Text style transfer for bias mitigation using masked language modeling**. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies: Student Research Workshop*, pages 163–171, Hybrid: Seattle, Washington + Online. Association for Computational Linguistics.
- Peter Willett. 2006. The porter stemming algorithm: then and now. *Program*.
- Mengzhou Xia, Anjalie Field, and Yulia Tsvetkov. 2020. **Demoting racial bias in hate speech detection**. In *Proceedings of the Eighth International Workshop on Natural Language Processing for Social Media*, pages 7–14, Online. Association for Computational Linguistics.
- Zhongbin Xie, Vid Kocijan, Thomas Lukasiewicz, and Oana-Maria Camburu. 2023. **Counter-GAP: Counterfactual bias evaluation through gendered ambiguous pronouns**. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 3761–3773, Dubrovnik, Croatia. Association for Computational Linguistics.

Eric Xing, Saranya Venkatraman, Thai Le, and Dongwon Lee. 2024. Alison: Fast and effective stylistic authorship obfuscation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19315–19322.

Mehdi Yazdani-Jahromi, AmirArsalan Rajabi, Ali Khodabandeh Yalabadi, Aida Tayebi, and Ozlem Ozmen Garibay. 2022. Distraction is all you need for fairness. *arXiv preprint arXiv:2203.07593*.

Marcos Zampieri, Shervin Malmasi, Preslav Nakov, Sara Rosenthal, Noura Farra, and Ritesh Kumar. 2019. [SemEval-2019 task 6: Identifying and categorizing offensive language in social media \(OffensEval\)](#). In *Proceedings of the 13th International Workshop on Semantic Evaluation*, pages 75–86, Minneapolis, Minnesota, USA. Association for Computational Linguistics.

Brian Hu Zhang, Blake Lemoine, and Margaret Mitchell. 2018. Mitigating unwanted biases with adversarial learning. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society*, pages 335–340.

A Appendix

A.1 DiffT

Here, re-sampling candidacy ($C(x)$) is computed using $C(x) \equiv |a(x)| - w_p m(x)$, where $a(x)$ is normalized p_{AE} rank (for example, for $p_{AE} \in \{0.9, 0.8, 0.7, 0.6, 0.5\}$, $a(x) \in \{1, 0.8, 0, -0.8, -1\}$. $a(x) = 0$ for median (Q2) of p_{AE} , $a(x) = 1$ for tweet with highest p_{AE} , $a(x) = -1$ for tweet with lowest p_{AE} and $w_p \in \{0.1, 0.32, 1, 3.2, 10\}$. The tweets are ranked based on $C(x)$ in descending order and is re-sampled based on the following:

- $p_{AE} > Q2$ and label = *non-toxic*, duplicate
- $p_{AE} > Q2$ and label = *toxic*, drop
- $p_{AE} < Q2$ and label = *toxic*, duplicate
- $p_{AE} < Q2$ and label = *non-toxic*, drop

We made the following changes to DiffT: We measure the bias as the difference of average false positive rate (FPR) for minority minus average FPR for majority. We also discarded the bias reduction threshold T . Instead, we keep iterating until the bias reduced. Also, we halt when the bias reduction changes to opposite direction and pick the best based on the closeness to 0. For example, we halt when bias reduced from 0.80 to -0.10 and pick the second trained model as the bias is closer to zero (0). Lastly, $m(x) = |p(\text{toxic}) - p(\text{non-toxic})|$

A.2 Model Configurations

A.2.1 MLP

The MLP toxicity classifier consist of an input layer, a hidden layer (size 128) and an output layer (size 2) with softmax function. It gives a binary prediction of *toxic* (positive class)/ *non-toxic* (negative class). The input text is stemmed using Porter Stemmer (Willett, 2006) and lower cased. Tokens⁸ are converted to real valued vectors using pre-trained Glove embedding of 300 dimension⁹. We randomly initialize the word embedding for words not in the embedding. The input is then post-padded or truncated to a maximum length which is the largest text length occurring at least 5 times in the dataset. We use sigmoid activation on the hidden layer output with a dropout of 0.5 in penultimate layer. The model learns optimal parameters by minimizing cross-entropy loss and the Adam optimizer with a learning rate of 1e-4, batch size of 64 for 50 epochs with early stopping (patience = 5). We also added L1/L2 regularization to handle overfitting. We use PyTorch for implementation using NVIDIA Tesla P100 PCIE (16GB) GPU. # of parameters ranged from 5 - 18 million depending on datasets. On average it took 1.5 days (GPU hours) for the cycle of training then debiasing the MLP per dataset.

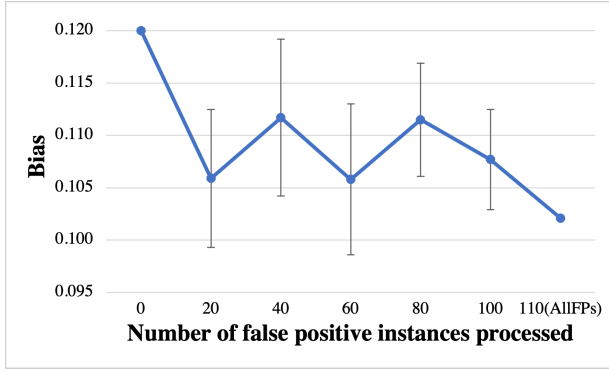
A.2.2 MLMs

We fine-tuned pre-trained MLMs classifiers for toxicity classification. This includes BERT (bert-base-uncased, bert-large-uncased) (Devlin et al., 2019), DistilBERT (distilbert-base-uncased) (Sanh et al., 2019) and RoBERTa-base (roberta-base) (Liu et al., 2019b). We use a batch size of 32, a maximum sequence length of 128, AdamW optimizer with a learning rate of 2e-5, and tune the models for ten epochs with early stopping (patience = 3). These hyper-parameters values are taken from the MLM papers. We implement these models using SimpleTransformers (Rajapakse, 2019).

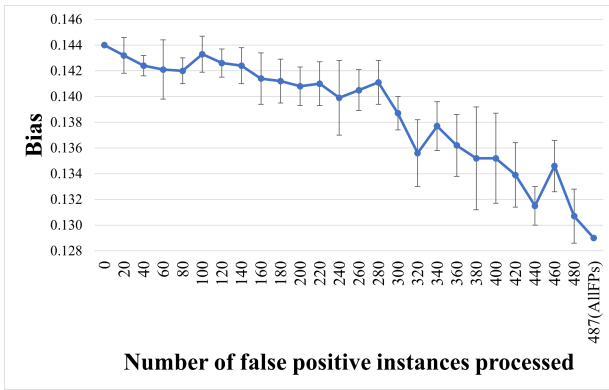
⁸https://www.tensorflow.org/api_docs/python/tf/keras/preprocessing/text/Tokenizer

⁹<https://nlp.stanford.edu/projects/glove/>

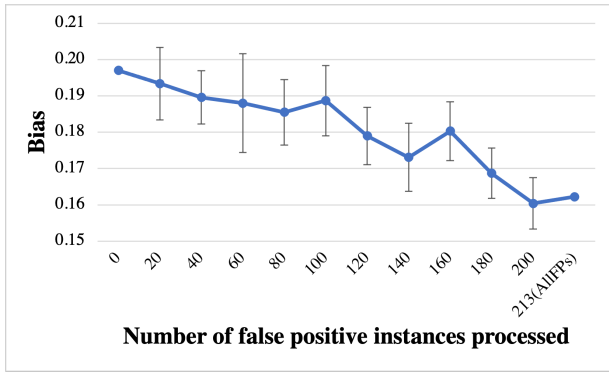
A.3 Sensitivity Analysis



(a) DWMW17



(b) FDCL18



(c) HatEval19

Figure 4: Bias on varying number of false positive instances (s) processed through obfuscation. Ten replications were done for each s . Here we use a single fixed seed initialized toxicity classifier network to explore the effect of varying s . Therefore, original bias (i.e. for $s=0$) is different from the ones reported in Tables 2-4 where we report averages over 10 runs. Vertical bars represent standard deviation across the ten samples. (MLP OBF_{TC})

Dataset	MLMs	Minority		Majority	
		AE	Hispanic (H)	White	Asian (AS)
DWMW17	DistilBERT	0.214	0.173	0.053	0.094
	BERT-base	0.197	0.180	0.054	0.062
	RoBERTa-base	0.229	0.176	0.059	0.073
	BERT-large	0.206	0.183	0.067	0.075
FDCL18	DistilBERT	0.198	0.166	0.031	0.016
	BERT-base	0.195	0.170	0.032	0.016
	RoBERTa-base	0.202	0.169	0.034	0.016
	BERT-large	0.189	0.163	0.030	0.017
HatEval19	DistilBERT	0.464	0.321	0.166	0.053
	BERT-base	0.435	0.346	0.175	0.060
	RoBERTa-base	0.419	0.288	0.164	0.077
	BERT-large	0.409	0.325	0.151	0.059

Table 5: Original FPR for minority and majority communities (MLMs)

MLMs	Approach	Majority		Minority		All		Bias	Sacrifices		
		FPR	F1	FPR	F1	FPR	F1		Bias	F1	FPR
DistilBERT	Original	0.073	0.934	0.194	0.898	0.134	0.916	0.121			
	OBFTC	-12.1	0.6	-7.1	0.6	-8.5	0.6	-4.0	●		
	OBFMIDAS	-30.4	0.9	-14.9	0.4	-19.1	0.7	-5.5	●		
	OBFNULI	-25.8	0.8	-14.2	0.9	-17.4	0.9	-7.2	●		
	OBFS _C	-25.9	1.1	-12.7	1.0	-16.3	1.0	-4.7	●		
BERT-base	Original	0.058	0.942	0.189	0.900	0.123	0.921	0.131			
	OBFTC	-5.0	0.0	-2.4	0.1	-3.0	0.0	-1.2	○		
	OBFMIDAS	4.2	0.0	-3.0	0.0	-1.3	0.0	-6.2	●		◇
	OBFNULI	-13.5	0.3	-6.8	-0.1	-8.4	0.1	-4.0	●		
	OBFS _C	-2.9	0.3	-4.8	0.4	-4.3	0.4	-5.6	●		
RoBERTa-base	Original	0.066	0.937	0.202	0.895	0.134	0.916	0.137			
	OBFTC	0.0	-0.4	-3.8	0.4	-2.9	0.0	-5.5	●		
	OBFMIDAS	-5.0	-0.1	-3.9	0.5	-4.2	0.2	-3.4	○		
	OBFNULI	0.0	-0.1	-3.7	0.9	-2.5	0.4	-6.2	●		
	OBFS _C	-15.3	0.5	-10.2	0.3	-11.5	0.4	-7.8	●		
BERT-large	Original	0.071	0.936	0.194	0.900	0.132	0.918	0.124			
	OBFTC	20.2	-0.6	2.5	-0.5	7.2	-0.5	-7.5	●		◇ ◇ ◇
	OBFMIDAS	6.2	-0.3	3.0	-0.6	3.9	-0.5	1.1	○		◇ ◇ ◇
	OBFNULI	-5.4	-0.1	-8.5	0.0	-7.7	0.0	-10.3	●		
	OBFS _C	0.0	-0.3	-5.5	-0.2	-4.0	-0.3	-8.8	●		

Table 6: (DWMW17 test set results). See Table 5 for *Original* FPR of each community and Table 2 for explanation of cell values and notations of *Sacrifices* column.

MLMs	Approach	Majority		Minority		All		Bias	Sacrifices		
		FPR	F1	FPR	F1	FPR	F1		Bias	F1	FPR
DistilBERT	Original	0.023	0.912	0.182	0.912	0.103	0.912	0.159			
	OBFTC	-4.8	-0.1	-4.9	0.2	-4.8	0.1	-4.9	●		
	OBFMIDAS	-4.3	0.2	-5.8	0.2	-5.6	0.2	-6.1	●		
	OBFNULI	-6.0	0.1	-5.7	0.1	-5.7	0.1	-5.6	●		
	OBFS _C	-9.0	0.0	-7.4	0.3	-7.6	0.2	-7.2	●		
BERT-base	Original	0.024	0.912	0.183	0.911	0.103	0.911	0.158			
	OBFTC	-8.2	0.1	-5.7	0.1	-6.0	0.1	-5.3	●		
	OBFMIDAS	-6.8	0.1	-5.0	0.0	-5.2	0.0	-4.7	●		
	OBFNULI	-12.1	0.1	-6.4	0.2	-7.1	0.1	-5.5	●		
	OBFS _C	-9.9	0.2	-6.0	0.2	-6.5	0.2	-5.4	●		
RoBERTa-base	Original	0.025	0.913	0.185	0.912	0.105	0.913	0.161			
	OBFTC	-9.4	0.1	-5.3	0.0	-5.8	0.0	-4.7	●		
	OBFMIDAS	-7.2	-0.1	-4.5	0.0	-4.9	0.0	-4.1	●		
	OBFNULI	-10.6	0.0	-4.4	0.1	-5.2	0.0	-3.5	○		
	OBFS _C	-7.6	-0.1	-6.1	0.2	-6.3	0.0	-5.9	●		
BERT-large	Original	0.024	0.909	0.176	0.912	0.100	0.911	0.152			
	OBFTC	-12.9	0.1	-6.7	0.1	-7.4	0.1	-5.7	●		
	OBFMIDAS	-9.9	0.3	-4.8	0.0	-5.4	0.1	-4.0	●		
	OBFNULI	0.0	0.1	-0.9	0.0	-0.7	0.0	-1.1	○		
	OBFS _C	-6.1	0.2	-2.3	0.0	-2.7	0.1	-1.7	○		

Table 7: (FDCL18 test set results). See Table 5 for *Original* FPR of each community and Table 2 for explanation of cell values and notations of *Sacrifices* column.

MLMs	Approach	Majority		Minority		All		Bias	Sacrifices		
		FPR	F1	FPR	F1	FPR	F1		Bias	F1	FPR
DistilBERT	Original	0.110	0.731	0.393	0.714	0.251	0.722	0.283			
	OBFTC	-5.6	-0.1	-7.6	0.5	-7.1	0.2	-8.3	●		
	OBFMIDAS	-1.3	1.8	-8.9	1.1	-7.2	1.5	-11.8	●		
	OBFNULI	-13.6	1.4	-11.9	1.0	-12.3	1.2	-11.3	●		
	OBFS _C	-7.8	1.1	-12.1	0.8	-11.2	0.9	-13.8	●		
BERT-base	Original	0.117	0.748	0.391	0.720	0.254	0.734	0.274			
	OBFTC	-13.5	-1.5	-13.4	0.8	-13.4	-0.4	-13.3	●		
	OBFMIDAS	5.0	-0.7	-2.0	0.0	-0.3	-0.4	-5.0	●		◇
	OBFNULI	-11.8	-1.4	-12.3	0.7	-12.2	-0.4	-12.5	●		
	OBFS _C	-3.4	-1.6	-5.8	0.6	-5.2	-0.5	-6.8	●		
RoBERTa-base	Original	0.121	0.743	0.353	0.727	0.237	0.735	0.233			
	OBFTC	4.2	1.1	-8.0	1.7	-4.9	1.4	-14.4	●		◇
	OBFMIDAS	0.0	-2.1	-4.5	1.3	-3.2	-0.4	-7.1	●		
	OBFNULI	-6.1	-1.7	-10.2	1.2	-9.2	-0.3	-12.4	●		
	OBFS _C	-6.7	0.6	-2.6	1.4	-3.6	1.0	-0.4	○		
BERT-large	Original	0.105	0.738	0.367	0.720	0.236	0.729	0.262			
	OBFTC	-3.7	-0.2	-8.9	0.5	-7.8	0.2	-11.0	●		
	OBFMIDAS	0.0	0.0	-7.7	1.0	-5.9	0.5	-10.9	●		
	OBFNULI	-0.9	-1.6	-8.4	0.2	-6.7	-0.7	-11.4	●		
	OBFS _C	0.0	-1.4	-1.7	0.6	-1.3	-0.4	-2.5	○		

Table 8: (HatEval19 test set results). See Table 5 for *Original* FPR of each community and Table 2 for explanation of cell values and notations of *Sacrifices* column.

007-classifier	Before obfuscation	After obfuscation
OBF _{TC}	why do y'all blame the woman for what she wears when it comes to rape? that shit is wrong.	why do y'all blame the woman for what she wears when it comes to rape? that envy is wrong.
	f**king machines mass vibratiuons	professional machines mass vibratiuons
OBF _{MIDAS}	Drita is that nicca ! lmao	Drita is that trials ! lmao
	b**ch you got plans! with me hoe!	redesigning you got plans! with me hoe!
OBF _{NULI}	let me know how to contribute to your legal counsel....sue their asses off for everything they have	let me know how to contribute to your legal counsel....sue their questionnaires off for everything they have
	I wanna be f**ked ! Find my name here	I wanna be agonized ! Find my name here
OBF _{SC}	What a f**king goal from Dele Alli	What a officialise goal from Dele Alli
	I can't f**king believe it.. but I want Roman to beat Taker.	I can't pineapple believe it.. but I want Roman to beat Taker.

Table 9: Text samples before and after obfuscation across datasets (**BERT-base toxicity classifier**). Random replacements are made using our greedy-select random-replace in order to trip the 007-classifiers and potentially produce a non-toxic instance for the toxicity classifier. We do not aim to maintain semantic coherence. Random replacements are efficient, as humans will never view the synthetic instances.